

- (c) Priya stores her website on a webserver.

To transmit the website data to the webserver she uses parallel duplex data transmission.

Describe how data is transmitted using parallel duplex data transmission.

.....

.....

.....

.....

.....

.....

.....

..... [4]

- 9 The contents of three binary registers have been transmitted from one computer to another. **Even parity** has been used as an error detection method.

The outcome after transmission is:

Register A and **Register C** have been transmitted **correctly**.

Register B has been transmitted **incorrectly**.

Complete the **Parity bit** for each register to show the given outcome.

	Parity bit							
Register A		0	1	0	0	1	0	1
Register B		1	0	0	0	0	0	1
Register C		1	0	0	0	0	1	1

[3]

- (b) The finance company is also worried about the security of the data stored on its servers.

The company has decided to encrypt the data to improve the security.

Describe how the data are encrypted.

.....

.....

.....

.....

.....

.....

.....

..... [4]

- 5 The three binary numbers in the registers given have been transmitted from one computer to another.

One binary number has been transmitted incorrectly. This can be identified by the use of a **Parity bit**.

Identify the binary number that has been transmitted **incorrectly**. Explain how you identified the incorrect binary number.

	Parity bit							
Register A	1	0	1	1	1	0	0	1
Register B	1	1	1	0	0	1	1	1
Register C	1	0	0	1	1	0	1	1

The binary number that has been transmitted incorrectly is in **Register**

Explanation

.....

.....

.....

.....

..... [4]

- 5 Sonia shares files with her friends. The method of data transmission she uses is half-duplex serial transmission.

(a) Describe how data is transmitted using half-duplex serial data transmission.

.....

.....

.....

.....

.....

.....

.....

..... [4]

(b) The system uses parity bits to check for errors during data transmission.

The outcome of four bytes after transmission is:

Byte 1	Byte 2	Byte 3	Byte 4
00110011	01010100	10110100	01110111

One of the bytes has been transmitted incorrectly.

Identify the byte that was transmitted incorrectly.

Byte

Explain how you identified the byte that was transmitted incorrectly.

.....

.....

.....

..... [4]

- 9 Maisey purchases a new router and attaches it to her computer. The connection she sets up uses duplex data transmission.

(a) Five statements are given about duplex data transmission.

Tick (✓) to show if the statement is **True** or **False**.

Statement	True (✓)	False (✓)
Duplex data transmission can be either serial or parallel		
Duplex data transmission is when data is transmitted both ways, but only one way at a time		
Duplex data transmission is always used to connect a device to a computer		
Duplex data transmission is when data is transmitted both ways at the same time		
Duplex data transmission automatically detects any errors in data		

[5]

- (b) Maisey's computer uses an integrated circuit (IC) for data transmission that sends multiple bits at the same time.

State whether the IC uses **serial** or **parallel** data transmission.

..... [1]

- (c) Maisey purchases a new printer and connects it to her computer using the USB port.

Explain **two** benefits of using a USB connection.

Benefit 1

.....

.....

.....

Benefit 2

.....

.....

.....

[4]

10 Data is valuable to a company.

- (a)** Companies use error detection methods to make sure that data is accurate.

One error detection method is the use of a check digit.

Explain what is meant by a check digit and how it is used to detect errors.

.....

.....

.....

.....

.....

.....

.....

..... [4]

0478/13 – NOVEMBER 2019

- (b)** The data stored by the library is archived at the end of each day. The archive is held on a server in the library office.

The data is encrypted with an 8-bit key. As some of the data is confidential, the library wants to make the encryption more secure.

- (i)** State how the library could make the encryption more secure.

.....

..... [1]

- (ii)** The term used to describe data before it is encrypted is plain text.

State the term used to describe encrypted data.

..... [1]

- 5 The contents of three binary registers have been transmitted from one computer to another. **Odd parity** has been used as an error detection method.

The outcome after transmission is:

- **Register A** and **Register B** have been transmitted **correctly**.
- **Register C** has been transmitted **incorrectly**.

Write the appropriate **Parity bit** for each register to show the given outcome.

	Parity bit							
Register A		0	1	0	0	0	1	1
Register B		0	0	0	0	1	1	1
Register C		0	0	0	0	0	1	1

[3]

2 A school network is used to transmit and store data about students.

(a) Different types and methods of transmission can be used to send data across the network.

Three descriptions about data transmission are given.

Tick (✓) **one Method** and tick (✓) **one Type** for each description.

Description	Method		Type		
	Serial (✓)	Parallel (✓)	Simplex (✓)	Half-duplex (✓)	Duplex (✓)
Data is sent down a single wire in a single direction only.					
Data is sent down multiple wires in both directions, at the same time.					
Data is sent down a single wire in both directions, but never at the same time.					

[3]

(b) Parity bits are used to help detect errors in data transmission. A parity bit is added to each binary value before transmission.

Three binary values are to be transmitted using **even** parity.

(i) Complete the parity bit that would be added to each binary value for even parity.

Binary value							Parity bit
1	1	0	0	1	1	1	
1	0	1	0	1	0	1	
0	1	1	0	1	0	0	

[3]

(ii) A number of errors occurred during data transmission.

State why a parity check may **not** detect transmission errors.

.....

..... [1]

(c) Data is encrypted using 128-bit symmetric encryption before it is transmitted.

(i) Explain what is meant by encryption.

.....
.....
.....
..... [2]

(ii) State how the strength of the encryption can be improved.

.....
..... [1]

0478/12 – JUNE 2020

3 A company collects and stores data about its customers. The data is stored on a server in the company's office.

The data is transmitted to cloud storage to create a back-up.

The data is encrypted using symmetric encryption before it is sent to the cloud storage.

(a) Describe how the data is encrypted.

.....
.....
.....
.....
.....
.....
.....
..... [4]

- 8 Leonard has a new laser printer to print letters for his business.

Leonard connects his printer to his computer using the USB port.

- (a) Give **three** benefits of using the USB port to connect the printer to the computer.

Benefit 1

.....

Benefit 2

.....

Benefit 3

.....

[3]

0478/13 – JUNE 2020

- 9 Four 7-bit binary values are being transmitted from one computer to another. An odd parity check is being used to check for errors in the binary values.

Write the correct **Parity bit** for each **7-bit binary value** to make sure it meets **odd** parity.

Parity bit	7-bit binary value
.....	0000011
.....	1000000
.....	0111111
.....	1010101

[4]

6 A Von Neumann model for a computer system contains several integrated circuits (IC).

(a) Parallel data transmission is used in an IC.

(i) Describe how data is transmitted using parallel data transmission.

.....
.....
..... [2]

(ii) Give **one** benefit of using parallel, rather than serial, data transmission.

.....
..... [1]

9 Victoria is entering data into a computer system. The data will be transmitted to cloud storage.

(a) An even parity check is used to check for errors in the binary values after transmission.

For each of the **7-bit binary values**, write the **Parity bit** that makes sure **even** parity is met.

7-bit binary value	Parity bit
1100010	
1001011	
0100010	
0010111	

[4]

(b) Identify **two** other error checking methods that could be used to check the binary values are correct after transmission.

Method 1

Method 2

[2]

- (c) A check digit is used to check whether data is correct when entered into the system.

Describe how a check digit can be used to make sure the data entered is correct.

.....

.....

.....

.....

.....

.....

.....

..... [4]

0478/12 – NOVEMBER 2020

- 2 Four 7-bit binary values are transmitted from one computer to another. A parity bit was added to each binary value creating 8-bit binary values. All the binary values have been transmitted correctly.

- (a) Tick (✓) to show whether an **Even** or an **Odd** parity check has been used for each binary value.

8-bit binary value	Even (✓)	Odd (✓)
11111111		
01100110		
01111011		
10000000		

[4]

- (b) The data will also be checked using a checksum.

Describe how a checksum can be used to check that the data has been transmitted correctly.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

0478/13 – NOVEMBER 2020

- 5 Arjun uses a scanner to create digital versions of some printed documents.

The scanner is attached to his computer using a USB connection.

- (a) Tick (✓) to show if the USB connection uses **Parallel** or **Serial** data transmission.

Describe your chosen method of data transmission.

Parallel ☐

Serial ☐

Description

.....

.....

.....

.....

[3]

(b) Give **three** benefits of a USB connection.

Benefit 1

.....

Benefit 2

.....

Benefit 3

.....

[3]

6 Elsa writes a paragraph in an examination about encryption.

There are several terms missing from the paragraph.

Complete the paragraph using the list of given terms. Not all terms may need to be used.

Some terms may be used more than once.

- algorithm
- alphanumeric
- cookie
- cypher
- key
- padlock
- plain
- word processed

The data is encrypted using a This is an that is used to scramble the data. The data before encryption is known as text. When the data has been encrypted it is known as text. To read the encrypted data it needs to be decrypted using a

[5]

7 **Four** 7-bit binary values are transmitted from one computer to another. A parity bit was added to each binary value creating 8-bit binary values. All the binary values have been transmitted correctly.

(a) Tick (✓) to show whether an **Even** or an **Odd** parity check has been used for each binary value.

8-bit binary value	Even (✓)	Odd (✓)
10000001		
10000010		
00101001		
00101000		

[4]

(b) A parity check may not always detect errors that have occurred in data transmission.

State why a parity check may not detect data transmission errors.

..... [1]

(c) Give **one** other error checking method that could be used to check for errors in data transmission.

..... [1]

(d) Electronic data about the final score for the match is transmitted to a central computer 30 kilometres away, using serial transmission.

(i) Explain why serial transmission is more appropriate than parallel transmission in this scenario.

.....

.....

.....

.....

.....

..... [3]

(ii) The data transmission is also half-duplex.

Describe half-duplex data transmission.

.....

.....

.....

..... [2]

(iii) The data transmission uses checksums.

Describe how checksums are used to detect errors in data transmission.

.....

.....

.....

.....

.....

..... [3]

(c) Julia uses a USB connection to transfer data onto her USB flash memory drive.

(i) One benefit of using a USB connection is that it is a universal connection.

State **two** other benefits of using a USB connection.

Benefit 1

.....

Benefit 2

.....

[2]

(ii) Identify the type of data transmission used in a USB connection.

..... [1]

8 Four 7-bit binary values are transmitted from one computer to another. A parity bit is added to each binary value creating 8-bit binary values. All the binary values are transmitted and received correctly.

(a) Identify whether each 8-bit binary value has been sent using odd or even parity by writing odd or even in the type of parity column.

8-bit binary value	Type of parity
01100100	
10010001	
00000011	
10110010	

[4]

(b) An error may **not** be detected when using a parity check.

Identify why an error may **not** be detected.

.....

..... [1]

(c) The data is sent using parallel half-duplex data transmission.

(i) Describe how data is sent using parallel half-duplex data transmission.

.....

.....

.....

.....

.....

.....

.....

..... [4]

(ii) State **two** drawbacks of using parallel data transmission.

Drawback 1

.....

Drawback 2

..... [2]

0478/13 – JUNE 2021

6 Four 7-bit binary values are transmitted from one computer to another. A parity bit is added to each binary value creating 8-bit binary values. All the binary values are transmitted and received correctly.

(a) Identify whether each 8-bit binary value has been sent using odd or even parity by writing odd or even in the type of parity column.

8-bit binary value	Type of parity
01111100	
10010000	
10011001	
00101001	

[4]

(b) The 8-bit binary value 10110001 is transmitted and received as 10110010

A parity check does **not** identify any errors in the binary value received.

State why the parity check does **not** identify any errors.

.....

..... [1]

(c) The data is sent using serial duplex data transmission.

(i) Describe how data is sent using serial duplex data transmission.

.....

.....

.....

.....

.....

.....

.....

..... [4]

(ii) State **one** drawback of using serial data transmission, rather than parallel data transmission.

.....

..... [1]

NOVEMBER 2021 (1)

- 3 Five statements are given about the error-checking methods checksum, check digit and parity check.

(a) Tick (✓) to show whether each statement applies to checksum, check digit or parity check. Some statements may apply to more than **one** error-checking method.

Statement	Checksum (✓)	Check digit (✓)	Parity check (✓)
uses an additional bit to create an odd or even number of 1s			
checks for errors on data entry			
compares two calculated values to see if an error has occurred			
will not detect transposition errors			
sends additional values when data is transmitted from a computer to another			

[5]

(b) Identify **one** other error-checking method.

..... [1]

NOVEMBER 2021 (2)

- 3 (a) Six statements are given about methods of data transmission.

Tick (✓) to show if each statement applies to serial simplex, parallel simplex, parallel half-duplex or serial duplex data transmission. Some statements may apply to more than **one** data transmission method.

Statement	Serial simplex (✓)	Parallel simplex (✓)	Parallel half-duplex (✓)	Serial duplex (✓)
bits are transmitted along a single wire				
data is transmitted in both directions				
it is only suitable for distances less than 5 metres				
bits from the same byte are transmitted one after the other				
data may not arrive in the correct sequence				
data is transmitted in both directions, but only one direction at a time				

[6]

- (b) A Universal Serial Bus (USB) connection can be used to transmit data from a mobile device to a computer.

Give **three** benefits of using a USB connection for this purpose.

Benefit 1

.....

Benefit 2

.....

Benefit 3

.....

[3]

3 Five statements are given about error-checking methods.

- (a) Tick (✓) to show whether each statement applies to Automatic Repeat reQuest (ARQ), check digit or checksum. Some statements may apply to more than **one** error-checking method.

Statement	ARQ (✓)	Check digit (✓)	Checksum (✓)
checks for errors on data entry			
uses a process of acknowledgement and timeout			
compares two calculated values to see if an error has occurred			
may resend data until it is confirmed as received			
checks for errors in data after transmission from a computer to another			

[5]

- (b) Identify **one** other error-checking method.

..... [1]

4 (c) The type of data transmission between the computer and the printer is serial half-duplex data transmission.

- (i) Describe how data is transmitted using serial half-duplex data transmission.

.....

.....

.....

.....

.....

.....

.....

..... [4]

- (ii) Explain why the data transmission needs to be half-duplex rather than simplex.

.....

.....

.....

..... [2]

- 9 A parity check is used to check for errors after transmission on the **four** given binary values.

All **four** values are transmitted and received correctly.

Identify whether each 8-bit binary value has been sent using odd or even parity by writing odd or even in the type of parity column.

Binary value	Type of parity
10011001	
01111110	
11100000	
00111001	

[4]

- (iii) Nadia's email service uses parity bits as a method of error detection during the transmission of data.

Describe how the parity bits are used to detect errors.

.....

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

- (iv) Identify and describe **one** other method of error detection used during the transmission of data.

Method

Description

.....

.....

.....

.....

.....

[4]

- 7 Umar is a manager in an office. The data that he uses on a daily basis is stored on a file server close to his computer.

Data is sent from his computer to the file server using parallel simplex data transmission.

- (a) Describe how the data is sent using parallel simplex data transmission.

.....

.....

.....

.....

.....

..... [3]

- (b) The file server is moved to another building that is 1 km away.

Explain why the parallel simplex data transmission method that Umar uses is no longer suitable.

.....

.....

.....

..... [2]

- (c) Checksum and Automatic Repeat reQuest (ARQ) are both used when transmitting the data from a computer to the file server.

Explain why checksum and ARQ are both used.

.....

.....

.....

.....

.....

..... [3]

5 A website allows users to purchase items.

Computer A sends a request for the homepage to the website's server.

(a) The request is sent using packet switching.

The structure of a packet of data has three elements. One element is the packet header.

(i) Identify **two** items of data contained in a packet header.

1

2 [2]

(ii) Identify the **two** other elements of a packet.

1

2 [2]

(b) Computer A needs to be directly connected to a router that is located in a different room.

(i) Tick (✓) **one** box to identify whether serial data transmission or parallel data transmission is more suitable for this connection.

Explain the reasons for your choice.

Serial data transmission

☐

Parallel data transmission

☐

Reasons for your choice

.....

.....

.....

.....

..... [3]

- (ii) The connection will also use full-duplex data transmission.

Define full-duplex data transmission.

.....

.....

.....

..... [2]

- (c) The data transmission will use parity checks.

- (i) The bytes need to be sent using an even parity byte check.

Complete the parity bit for each byte.

	Parity bit							
Byte A		1	1	0	0	0	1	1
Byte B		0	0	0	0	0	0	0

[2]

- (ii) A parity block check can be used instead of a parity byte check.

Explain how a parity block check might detect an error in transmission that would **not** be detected by a parity byte check.

.....

.....

.....

..... [2]

- (iii) The data was sent using an even parity block check. One of the bits has been transmitted incorrectly.

	Parity bit	Bit 1	Bit 2	Bit 3	Bit 4	Bit 5	Bit 6	Bit 7
Byte 0	1	1	1	0	1	0	0	0
Byte 1	0	0	1	0	0	1	0	0
Byte 2	1	0	1	1	0	0	0	1
Byte 3	1	1	0	0	1	1	1	1
Byte 4	1	0	1	0	0	0	1	0
Byte 5	0	0	0	0	0	0	0	0
Byte 6	0	1	1	1	1	0	0	0
Parity byte	0	1	1	0	1	0	1	0

Identify the bit number and the byte number of the incorrect bit.

Bit number

Byte number

[2]

(c) The student sends the sound file to a friend. The file is transmitted across a network that uses packet switching.

(i) Identify **two** pieces of data that would be included in the header of each packet.

1

2

[2]

(ii) Explain how the file is transmitted using packet switching.

.....

.....

.....

.....

.....

.....

.....

.....

.....

..... [5]

- 4 A student uses a mobile phone to take photographs for a school project.

The student needs to transmit the photographs to their computer. They could use serial data transmission or parallel data transmission to transmit the photographs.

- (a) (i) Describe how the photographs would be transmitted using serial data transmission.

.....

.....

.....

..... [2]

- (ii) Give **two** benefits of transmitting the photographs using serial data transmission.

1

.....

2

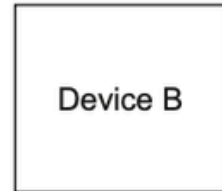
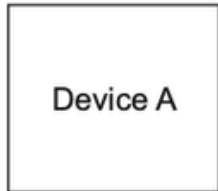
..... [2]

- (iii) State **one** benefit of the student using parallel data transmission instead of serial data transmission.

.....

..... [1]

- 5 Complete and annotate the diagram to demonstrate how packet switching is used to transmit data across a network, including the use of routers, from Device A to Device B.



[4]

4 Data is transmitted between a computer and a printer.

- (a) The data is transmitted one bit at a time down a single wire. The computer can transmit data to the printer and the printer can transmit data to the computer, using the same connection.

Circle the **two** data transmission methods that will transmit data in this way.

parallel full-duplex

parallel half-duplex

parallel simplex

serial full-duplex

serial half-duplex

serial simplex

[2]

- (b) An odd parity check is used to detect errors in the data transmission.

Explain how the odd parity check detects errors.

.....

.....

.....

.....

.....

.....

.....

.....

..... [4]

- (c) Another error detection method sends the data from the computer to the printer, then a copy of the data received is sent back from the printer to the computer. The two sets of data are compared to see if they match.

State the name of this type of error detection method.

..... [1]

7 Complete the statements about data packets and packet switching.

Use the terms from the list.

Some of the terms in the list will **not** be used. You should only use a term once.

destination address

first

footer

header

last

main data

packets

payload

routers

servers

trailer

Data is broken down into packets. A data packet has a packet
that contains the packet number and the

Each packet could take a different path from the sender to the receiver; this is controlled by
.....

Packets may arrive out of order. Once the packet has arrived,
the packets are reordered.

[4]

6 The table contains descriptions about data transmission methods.

Complete the table by identifying which data transmission methods are described.

Data transmission method	Description
.....	Data is transmitted down a single wire, one bit at a time, in one direction only.
.....	Data is transmitted down multiple wires, multiple bits at a time, in both directions, but only one direction at a time.
.....	Data is transmitted down a single wire, one bit at a time, in both directions at the same time.
.....	Data is transmitted down multiple wires, multiple bits at a time, in one direction only.

[4]

5 Errors can occur when data is transmitted.

(a) Give **one** reason an error may occur when data is transmitted.

.....
..... [1]

(b) Some error detection methods use a calculated value to check for errors.

Tick (✓) **one** box to show which error detection method does **not** use a calculated value to check for errors.

- | | | |
|----------|--------------|--------------------------|
| A | Check digit | ☐ |
| B | Checksum | ☐ |
| C | Echo check | ☐ |
| D | Parity check | ☐ |

[1]

(c) An automatic repeat request (ARQ) can be used to make sure that data is received free of errors. It can use a positive or negative acknowledgement method to do this.

Explain how an ARQ operates using a positive acknowledgement method.

.....
.....
.....
.....
.....
.....
.....
.....
.....
.....
..... [5]